

1. Write down the sequence of socket library functions invoked on the server side before a server is ready for communicating with the client using TCP. Give the important parameters and return values of these functions and write one sentence each about these functions.

2. A set of nodes form a connected graph where each edge denotes a neighbor relation. Two nodes with an edge between them are said to be one hop apart. Two hop neighbors are connected through an intermediate node. Design a protocol that permits a node to generate the list of all its 1-hop and 2-hop neighbors. Clearly state the message structures and actions of all the entities involved.

3. A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2 Mbps. What is the queuing delay for the packet?

4. Internet routing uses hop-count as a metric (distance) for computing shortest path. Since many applications need to minimize delay in the packet discovery, delay as a metric may be a better choice. What do you think are the reasons for choosing hop-count and not delay as the metric in Internet routing.

5. TCP computes the time-out for each packet as a function of estimated RTT and variance of RTT. Other possible choices may only be a function of estimated RTT. say $k \cdot \langle \text{RTT} \rangle$ or $\langle \text{RTT}_{\text{max}} \rangle$ etc. Write down the expression used by the TCP for the time-out and give reasons why such an expression is chosen instead of other choices mentioned above.

6. Describe the difference between pure and slotted ALOHA protocol. Why does the slotted ALOHA perform with twice the efficiency of pure ALOHA?

7. Explain the functioning of a MAC level switch. How does it lead to improvement of performance in a high load LAN?

8. Describe the important changes in IPv6 compared to IPv4 with the reason for each of the changes.

9. Describe the main elements of CSMA/CD protocol for wireless MAC. CSMA/CD is not usable on a wireless channel. Explain the reasons.